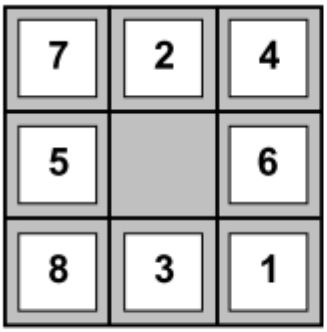
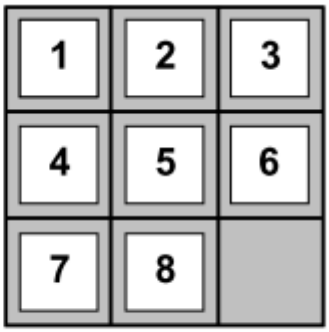
Exercises for Search

1. Formulate the 8-puzzle problem by specifying the states, actions, goal test and path costs.

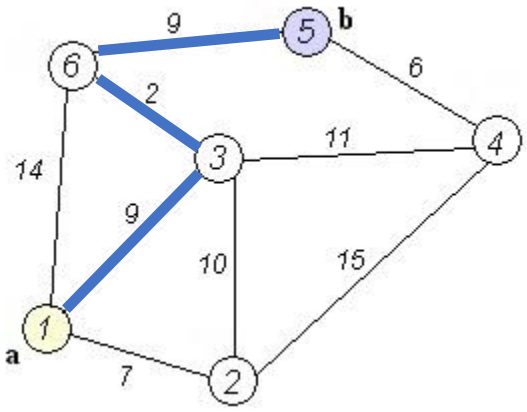


Start State



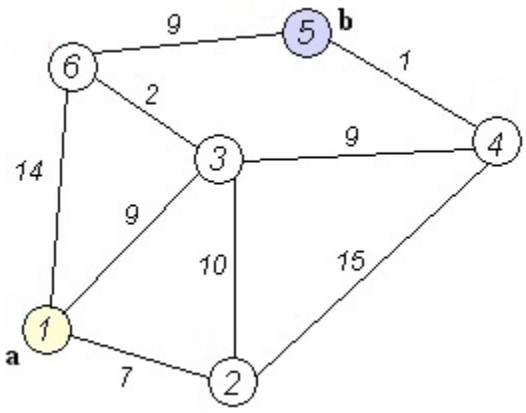
Goal State

2. For the route planning problem shown in the figure, six cities are labelled as 1-6, and their connectivity and distances are given on the edges in the graph. If we start from city 1 and the destination is city 5, apply the following search algorithms to find a path. Provide details of the node expansion and the found path from start to goal.
   1) Breadth-first search
   2) Depth-first search
   3) Uniform-cost search



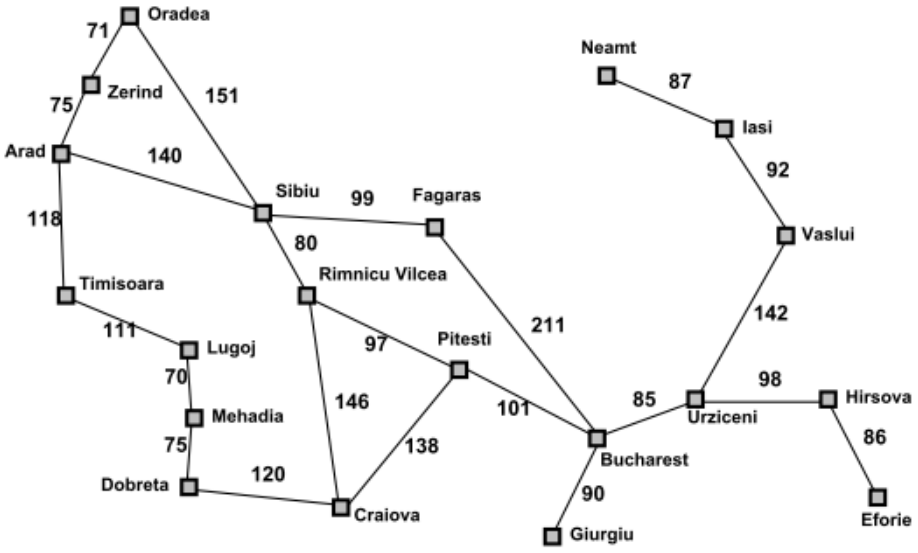
3. In the previous problem, if cost(3,4)=9 and cost(4,5)=1, apply the same search methods:
   1) Breadth-first search
   2) Depth-first search
   3) Uniform-cost search

   Would the search process and the final path be different for each method? Why?



4. Revise the Romania problem in the lecture. The start is Arad and the destination is Bucharest. If the heuristic function is defined by the straight-line distance to Bucharest as provided in the figure below, apply the following search algorithms to find a path. Provide details of node expansion, the found path and its total distance.
   1) Greedy best-first search
   2) A* search

   Please note that the straight-line distance for Pitesti in the table has been changed from 100 (as in the book) to 98.



Straight−line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	176
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

5. Discuss the conditions for optimality of A* search. In the following graph, the true distances between nodes are shown on the edges (in black), and the heuristic distance to G are shown on top of the node (in blue). Explain if these heuristic values are (1) admissible and (2) consistent. Apply the A* search to find a path from S to G, and verify if it is on the optimal path when first time reaching G.

